

Machine Learning

Python assignement-05

Date: 11th may 2026

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Questions: -

Q1. Print elements from a list present at odd index position.

Input: [10,20,30,40,50,60,70,80,90,100]

Q2. Count total no. of digit in a number.

Input: 75869

Q3. Reverse a integer no.

Input: 75869

Q4. Find largest and smallest digit in a given number.

Input:75869

Q5. Find factorial of a no. 5.

Q6. Check Armstrong Number 153.

Q7. Print right angled triangle pattern using a loop.

Input: 5

Q8. Using above Question print decreasing pattern.

Q9. Print alternate no. pattern.

Input: range 1 to 20

Q10. Find common elements (Intersection) using loop.

Given Input:

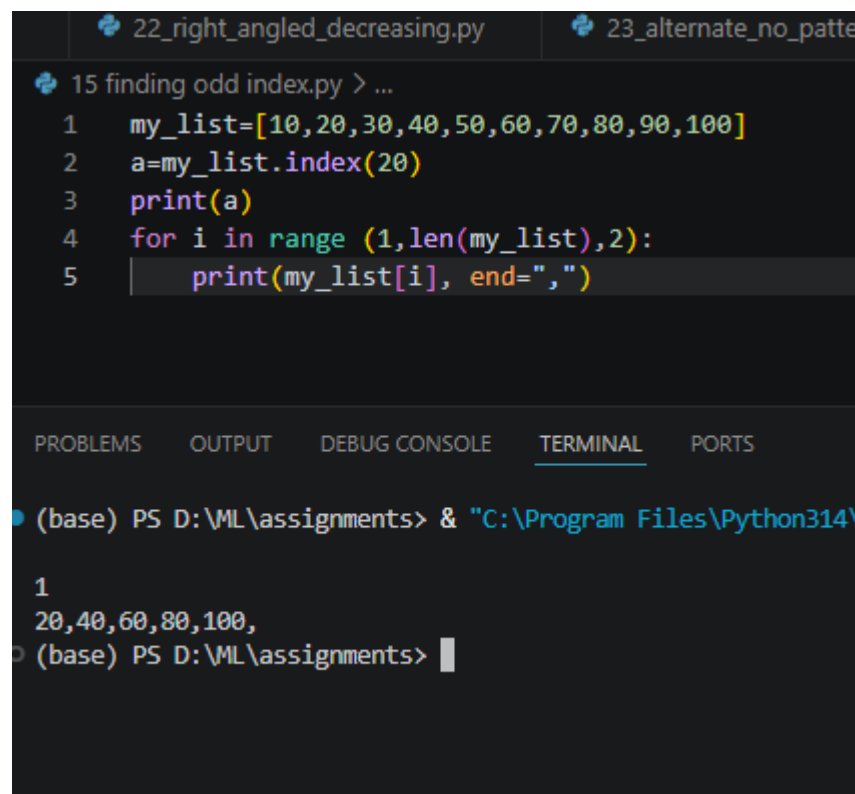
List_a=[1,2,3,4,5]

List_b=[4,5,6,7,8]

Solutions: -

Q1. Print elements from a list present at odd index position.

Input: [10,20,30,40,50,60,70,80,90,100]



The screenshot shows a code editor with two tabs: '22_right_angled_decreasing.py' and '23_alternate_no_patte'. The active tab contains a Python script with the following code:

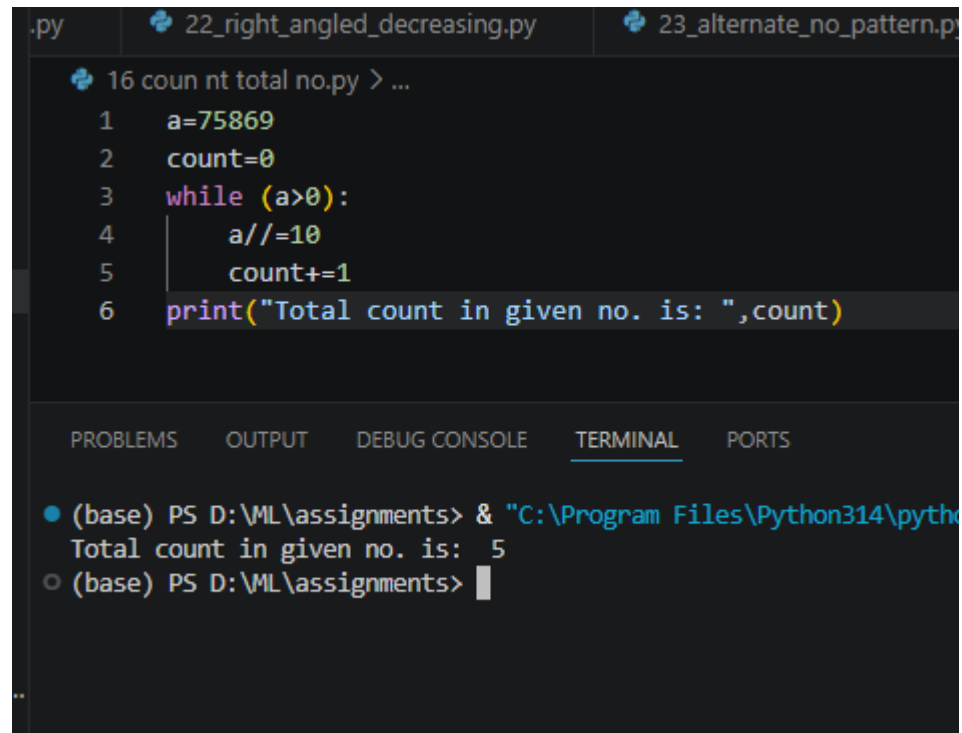
```
15 finding odd index.py > ...
1  my_list=[10,20,30,40,50,60,70,80,90,100]
2  a=my_list.index(20)
3  print(a)
4  for i in range (1,len(my_list),2):
5      print(my_list[i], end=",")
```

Below the code editor, there is a terminal window with tabs for 'PROBLEMS', 'OUTPUT', 'DEBUG CONSOLE', 'TERMINAL', and 'PORTS'. The 'TERMINAL' tab is active, showing the command prompt output:

```
(base) PS D:\ML\assignments> & "C:\Program Files\Python314\
1
20,40,60,80,100,
(base) PS D:\ML\assignments>
```

Q2. Count total no. of digit in a number.

Input: 75869



The image shows a code editor with two tabs: '22_right_angled_decreasing.py' and '23_alterate_no_pattern.py'. The active tab is '22_right_angled_decreasing.py', which contains the following Python code:

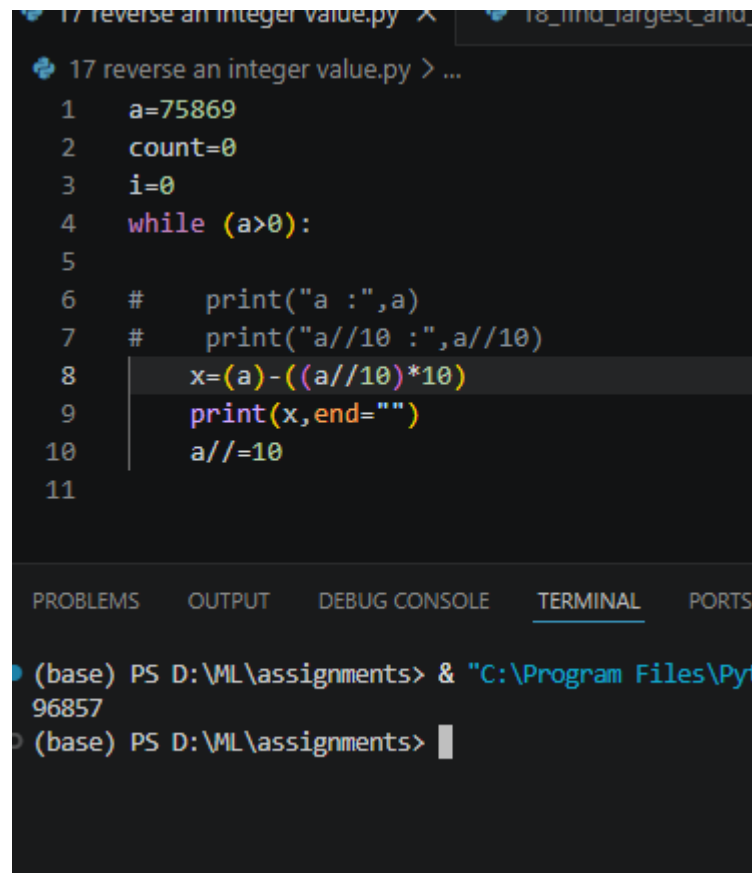
```
16 count total no.py > ...
1  a=75869
2  count=0
3  while (a>0):
4      a//=10
5      count+=1
6  print("Total count in given no. is: ",count)
```

Below the code editor is a terminal window with the following output:

```
(base) PS D:\ML\assignments> & "C:\Program Files\Python314\python.exe" 22_right_angled_decreasing.py
Total count in given no. is: 5
(base) PS D:\ML\assignments>
```

Q3. Reverse an integer no.

Input: 75869



The image shows a code editor with a Python script to reverse an integer. The script is named '17 reverse an integer value.py'. It contains the following code:

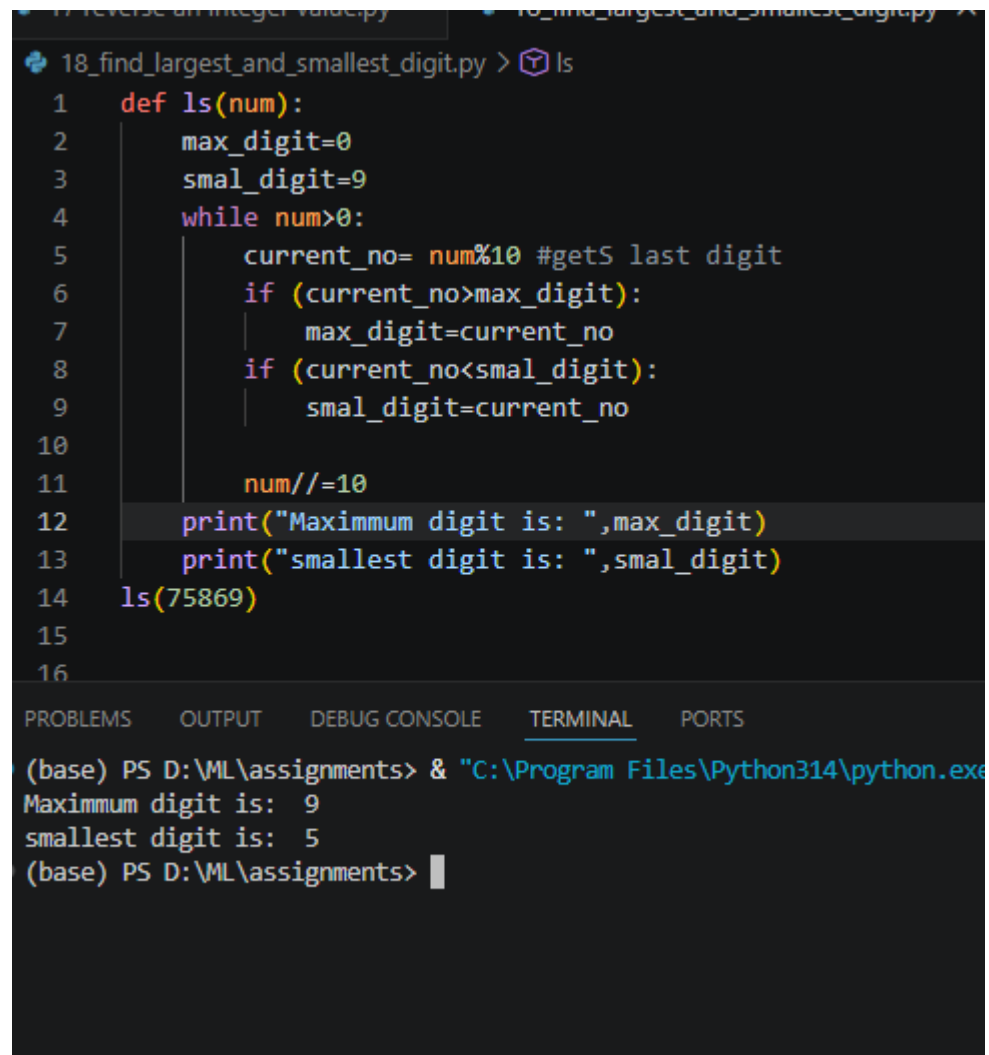
```
1 a=75869
2 count=0
3 i=0
4 while (a>0):
5
6     # print("a :",a)
7     # print("a//10 :",a//10)
8     x=(a)-((a//10)*10)
9     print(x,end="")
10    a//=10
11
```

Below the code editor, there is a terminal window with the following output:

```
(base) PS D:\ML\assignments> & "C:\Program Files\Python\Python39\python.exe" 17 reverse an integer value.py
96857
(base) PS D:\ML\assignments>
```

Q4. Find largest and smallest digit in a given number.

Input:75869



The image shows a code editor with a Python script named `18_find_largest_and_smallest_digit.py`. The script defines a function `ls(num)` that iterates through the digits of a number to find the maximum and minimum values. The code is as follows:

```
1 def ls(num):
2     max_digit=0
3     smal_digit=9
4     while num>0:
5         current_no= num%10 #getS last digit
6         if (current_no>max_digit):
7             max_digit=current_no
8         if (current_no<smal_digit):
9             smal_digit=current_no
10
11         num//=10
12     print("Maximmum digit is: ",max_digit)
13     print("smallest digit is: ",smal_digit)
14 ls(75869)
15
16
```

Below the code editor, the terminal output is shown. It displays the execution of the program, which correctly identifies the maximum digit as 9 and the minimum digit as 5 for the input 75869.

```
(base) PS D:\ML\assignments> & "C:\Program Files\Python314\python.exe"
Maximmum digit is: 9
smallest digit is: 5
(base) PS D:\ML\assignments>
```

Q5. Find factorial of a no. 5.

```
19_factorial_find.py > ...
1  n=5
2  a=n
3  for i in range(1,a):
4      a*=i
5
6  print(f"Factorial of {n} is: ",a)
7

PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PO

● (base) PS D:\ML\assignments> & "C:\Program Files
Factorial of 5 is: 120
○ (base) PS D:\ML\assignments> |
```

Q6. Check Armstrong Number 153.

Input:	
<pre>20_check_armstrong_no.py > Ⓜ armstrong 1 def armstrong(given_num): 12 #checking armstrong no. 13 sum=0 14 for i in range (count): 15 x=num1%10 16 print(x) 17 a=x**count 18 num1//=10 19 print(f"{'cube' if count==3 else 'power four'} for {i} digit",a) 20 sum+=a 21 print(sum) 22 print("\n\n") 23 if sum==given_num: 24 print(f"{given_num} is armstrong number") 25 else: 26 print(f"{given_num} is not armstong number") 27 armstrong(153) 28 armstrong(9999) 29 armstrong(given_num=9474)</pre>	

Output:

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

• (base) PS D:\ML\assignments> & "C:\Program Files\Python\Python38\python.exe" .\armstrong.py
total number of digits in given number:  3
3
cube for 0 digit 27
5
cube for 1 digit 125
1
cube for 2 digit 1
153

153 is armstrong number
total number of digits in given number:  4
9
power four for 0 digit 6561
9
power four for 1 digit 6561
9
power four for 2 digit 6561
9
power four for 3 digit 6561
26244

9999 is not armstrong number
total number of digits in given number:  4
4
power four for 0 digit 256
7
power four for 1 digit 2401
4
power four for 2 digit 256
9
power four for 3 digit 6561
9474

9474 is armstrong number
• (base) PS D:\ML\assignments>
```

Q7. Print right angled triangle pattern using a loop.

Input: 5

```
21_right_angles_pattern.py > ...
1  n=5
2  for i in range(0,n+1):
3      # print("\n")
4      for j in range(0,i):
5          print("*",end=" ")
6      print()

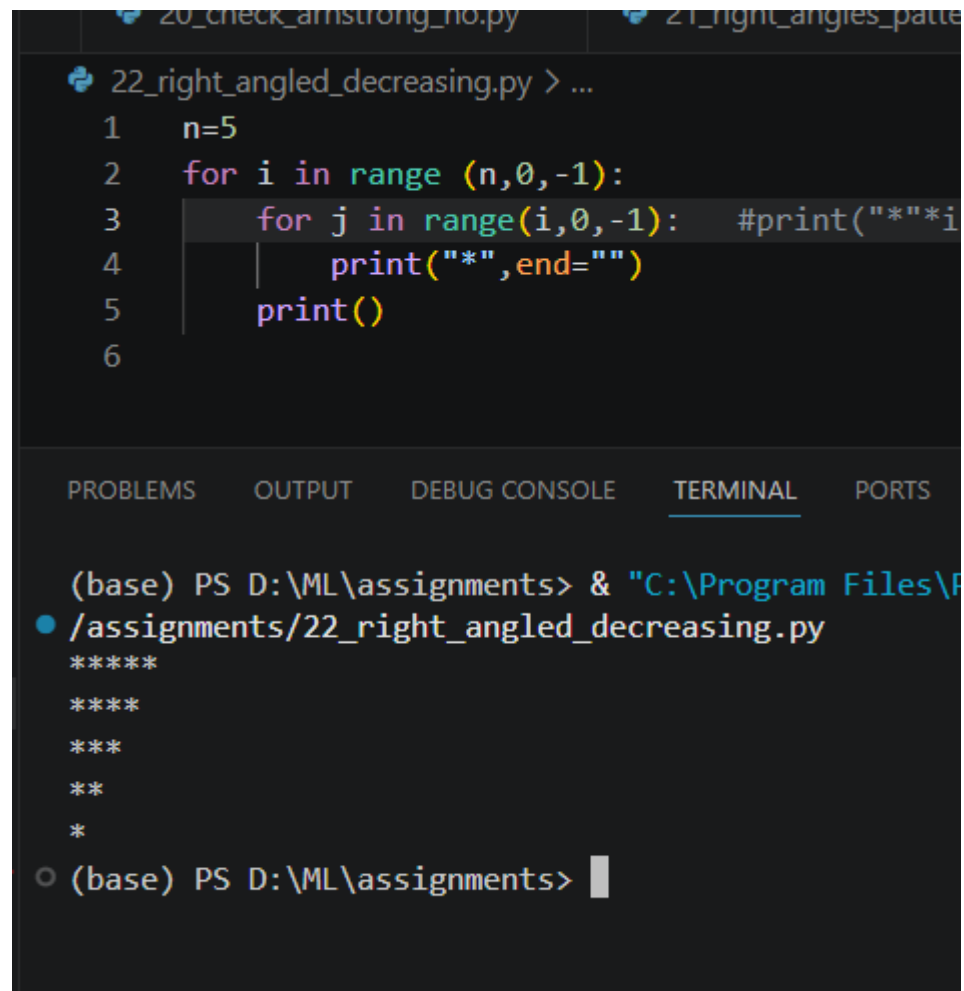
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL

(base) PS D:\ML\assignments> & "C:\Program Files\Python\Python38-32\python.exe" D:\ML\assignments/21_right_angles_pattern.py

*
* *
* * *
* * * *
* * * * *

(base) PS D:\ML\assignments>
```

Q8. Using above Question print decreasing pattern.



The image shows a code editor window with a file named `22_right_angled_decreasing.py`. The code is as follows:

```
1 n=5
2 for i in range (n,0,-1):
3     for j in range(i,0,-1): #print("***i
4         print("*",end="")
5     print()
6
```

Below the code editor is a terminal window. The terminal shows the command to run the script and its output:

```
(base) PS D:\ML\assignments> & "C:\Program Files\F
● /assignments/22_right_angled_decreasing.py
*****
****
***
**
*
○ (base) PS D:\ML\assignments>
```

Input: range 1 to 20

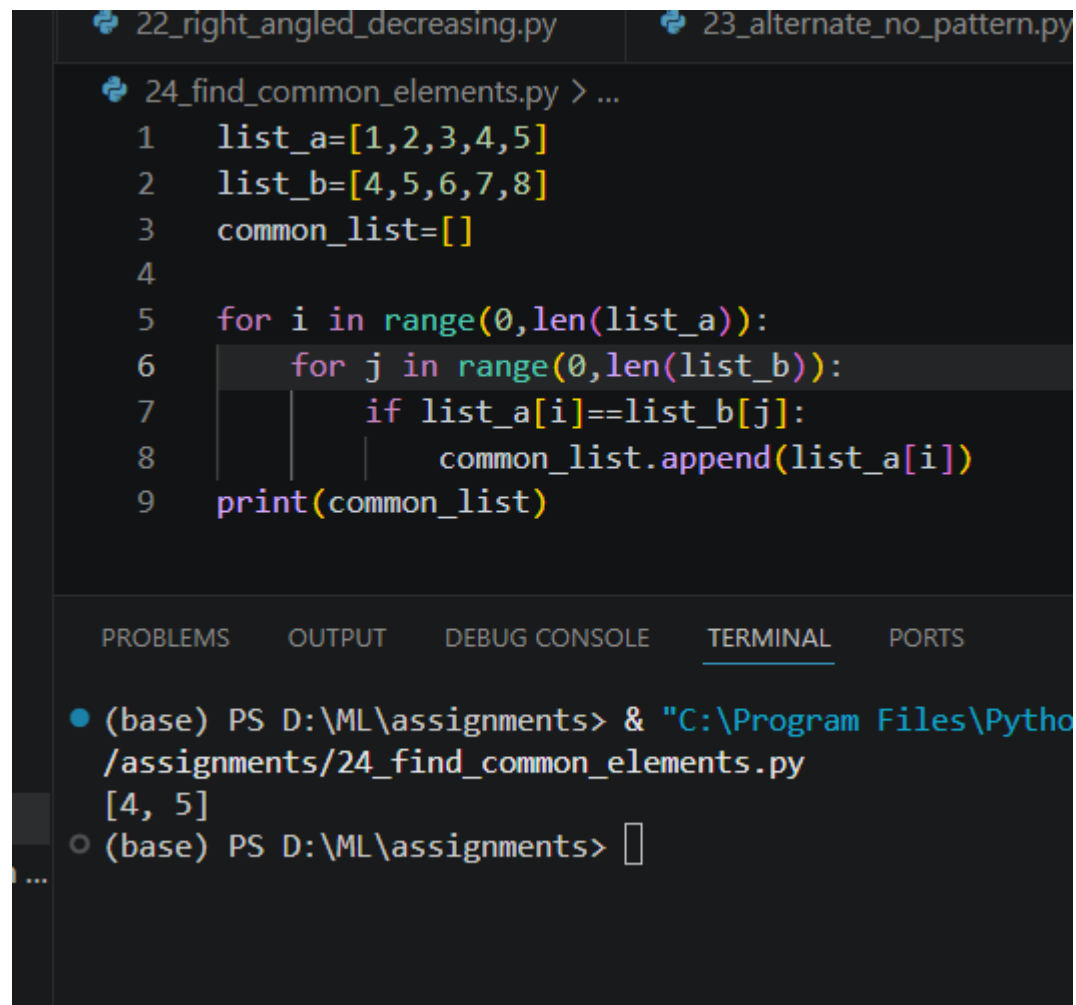
[illegible]

Q10. Find common elements (Intersection) using loop.

Given Input:

List_a=[1,2,3,4,5]

List_b=[4,5,6,7,8]



```
22_right_angled_decreasing.py 23_alternate_no_pattern.py
24_find_common_elements.py > ...
1 list_a=[1,2,3,4,5]
2 list_b=[4,5,6,7,8]
3 common_list=[]
4
5 for i in range(0,len(list_a)):
6     for j in range(0,len(list_b)):
7         if list_a[i]==list_b[j]:
8             common_list.append(list_a[i])
9 print(common_list)

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
● (base) PS D:\ML\assignments> & "C:\Program Files\Python\python.exe" D:\ML\assignments/24_find_common_elements.py
[4, 5]
○ (base) PS D:\ML\assignments> 
```